

Week 10 Discussion: Wiener Processes, Itô's Lemma, & Black–Scholes–Merton

Main focus (25min): (i) generalized Wiener processes + correlation; (ii) Itô processes + GBM/lognormality; (iii) universal application of BS equation to GBM and pricing formulas.

Module A) Generalized Wiener Processes & Correlation

Generalized Wiener (Brownian motion with drift):

$$X_t = X_0 + \mu t + \sigma B_t \quad \Longleftrightarrow \quad dX_t = \mu dt + \sigma dB_t,$$

so increments satisfy $X_{t+\Delta t} - X_t \sim N(\mu\Delta t, \sigma^2\Delta t)$.

Two processes and correlation: If

$$dX_1 = \mu_1 dt + \sigma_1 dB_1, \quad dX_2 = \mu_2 dt + \sigma_2 dB_2,$$

and $\text{Corr}(dB_1, dB_2) = \rho$, then

$$d(X_1 + X_2) = (\mu_1 + \mu_2) dt + (\sigma_1 dB_1 + \sigma_2 dB_2),$$

with **variance rate**

$$\sigma_{sum}^2 = \sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2 \quad (\text{so } \rho = 0 \Rightarrow \sigma_{sum}^2 = \sigma_1^2 + \sigma_2^2).$$

Practice A (Textbook 14.3): Sum of generalized Wiener processes

Variables X_1 and X_2 follow generalized Wiener processes with drift rates μ_1, μ_2 and variances σ_1^2, σ_2^2 . What process does $X_1 + X_2$ follow if:

1. the changes in X_1 and X_2 in any short interval are uncorrelated?
2. there is correlation ρ between the changes in any short interval?

Module B) Itô's Lemma & Geometric Brownian Motion (GBM)

Itô process: $dX_t = K_t dt + H_t dB_t$.

Itô's lemma (template): for $Z_t = g(t, X_t)$ with $g \in C^2$,

$$dZ_t = \left(g_t + g_x K_t + \frac{1}{2} g_{xx} H_t^2 \right) dt + g_x H_t dB_t.$$

GBM stock model: $dS_t = \mu S_t dt + \sigma S_t dB_t$.

Key identity (log trick):

$$d(\ln S_t) = \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dB_t \quad \Rightarrow \quad \ln\left(\frac{S_T}{S_0}\right) \sim N\left(\left(\mu - \frac{1}{2}\sigma^2\right)T, \sigma^2 T\right).$$

So S_T is **lognormal**.

Practice B (Textbook 14.6): Does a portfolio follow GBM?

Stock A and stock B both follow geometric Brownian motion. Changes in any short interval of time are uncorrelated with each other. Does the value of a portfolio consisting of *one share of A and one share of B* follow geometric Brownian motion? Explain briefly.

Module C) Risk-Neutrality, Universal BS Equation & Pricing Formulas

Risk-neutral modeling (no dividends): replace μ by $r \Rightarrow dS_t = rS_t dt + \sigma S_t dB_t^Q$.

Black–Scholes Formulas (European call/put, no dividends): with $\tau = T - t$,

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau},$$

$$C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \quad P = Ke^{-r\tau}\Phi(-d_2) - S\Phi(-d_1).$$

Mini warm-up (Textbook 15.2): If annual volatility is $\sigma = 30\%$, then over 1 trading day ($\Delta t \approx 1/252$ year), $\text{SD}(\%\Delta S) \approx \sigma\sqrt{\Delta t} = 0.30/\sqrt{252} \approx 1.9\%$.

Applicability of the BS Equation: The Black–Scholes partial differential equation (PDE) applies to **ALL** derivative securities whose underlying asset follows a Geometric Brownian Motion (GBM).

Practice C (Textbook 15.16): BSM and Put–Call Parity

Show that the Black–Scholes–Merton formulas for call and put options satisfy put–call parity.